

Hongjie Chen

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Research Interests

Privacy-preserving machine learning, particularly differential privacy
High-dimensional statistical estimation

Education

ETH Zürich

PH.D. COMPUTER SCIENCE

2022 - Present

Advised by David Steurer

ETH Zürich

M.S. COMPUTER SCIENCE (WITH DISTINCTION)

2019 - 2022

Advised by David Steurer

Thesis: On the Spreadness Property and Lower Bounds for Robust Regression

Shanghai Jiao Tong University

B.E. COMPUTER SCIENCE AND ENGINEERING

2015 - 2019

Advised by Chihao Zhang

Thesis: An Empirical Study on Mixing Time of Markov Chains

Professional Experience

Apple Machine Learning Research

RESEARCH INTERN

Hosted by Hilal Asi

Cupertino, USA

May - Sep 2025

Caltech

VISITING UNDERGRADUATE RESEARCH INTERN

Hosted by Anima Anandkumar and Kamyar Azizzadenesheli

Pasadena, USA

Jul - Sep 2018

Publications

On efficient robust regression with subquadratic samples. Deeksha Adil, Jaroslaw Blasiok, Hongjie Chen, Deepak Narayanan. *COLT* 2026.

Fast and optimal algorithms for private hypothesis selection. Hilal Asi, Hongjie Chen. *ICML* 2026.

Improved robust estimation for Erdős-Rényi graphs: the sparse regime and optimal breakdown point. Hongjie Chen, Jingqiu Ding, Yiding Hua, Stefan Tiegel. *NeurIPS* 2025.

Outlier-robust mean estimation near the breakdown point via sum-of-squares. Hongjie Chen, Deepak Narayanan, David Steurer. *SODA* 2025.

Private edge density estimation for random graphs: optimal, efficient and robust. Hongjie Chen, Jingqiu Ding, Yiding Hua, David Steurer. *NeurIPS* 2024 (*spotlight*).

Private graphon estimation via sum-of-squares. Hongjie Chen, Jingqiu Ding, Tommaso d'Orsi, Yiding Hua, Chih-Hung Liu, David Steurer. *STOC* 2024.

Private estimation algorithms for stochastic block models and mixture models. Hongjie Chen, Vincent Cohen-Addad, Tommaso d'Orsi, Alessandro Epasto, Jacob Imola, David Steurer, Stefan Tiegel. *NeurIPS* 2023 (*spotlight*).

On the well-spreadness property and its relation to linear regression. Hongjie Chen, Tommaso d'Orsi. *COLT* 2022.

Honors & Awards

ETH Medal for Outstanding Master's Thesis

2022

Rongchang Scholarship, Shanghai Jiao Tong University

2017, 2018

Zhiyuan Scholarship, Shanghai Jiao Tong University

2016, 2017

Teaching & Service

Presenting Theoretical Computer Science, ETH Zürich,	Teaching Assistant	2026
Algorithms and Data Structures, ETH Zürich,	Head Teaching Assistant	2024
Algorithms and Data Structures, ETH Zürich,	Head Teaching Assistant	2023
Optimization for Data Science, ETH Zürich,	Teaching Assistant	2023
Presenting Theoretical Computer Science, ETH Zürich,	Teaching Assistant	2023
Algorithms and Data Structures, ETH Zürich,	Teaching Assistant	2022

Conference reviewing: STOC, NeurIPS, ITCS, ICML, COLT